



infinITY Online Camp JEE (Adv) 2024

Achieve Your
DREAM
IIT in
2024

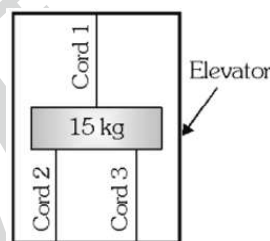
NLM & FRICTION

SUBJECT: PHYSICS

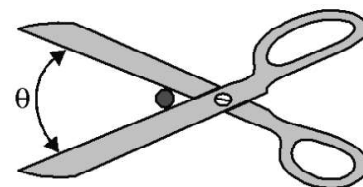
DATE: 16-03-2024

SINGLE CORRECT TYPE (+3, -1)

1. A block of mass $m = 15\text{ kg}$ is suspended in an elevator with the help of three identical light elastic cords (spring constant $k = 100\text{ N/m}$ each) attached vertically. One of them, cord 1 is tied to the ceiling of the elevator and the other two cords 2 and 3 are tied to the elevator floor as shown in the figure. When the elevator is stationary the tension force in each of the lower cords is $T = 7.5\text{ N}$. Take $g = 10\text{ m/s}^2$. Now the elevator starts moving with some acceleration

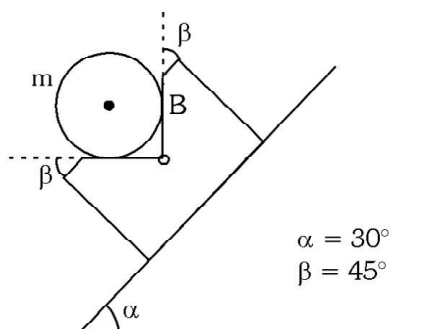


- (A) if acceleration of lift is 1.5 m/s^2 upward then displacement of the block w.r.t. elevator upto new equilibrium position is 17.5 cm downward and tension in the cord 1 is 172.5 N
- (B) if acceleration of lift is 1 m/s^2 upward then displacement of the block w.r.t. elevator upto new equilibrium position is 5 cm downward and tension in the cord 2 is 5 N
- (C) if acceleration of lift is 2 m/s^2 upward then displacement of the block w.r.t. elevator upto new equilibrium position is 7.5 cm downward and tension in the cord 3 is 0 N
- (D) if acceleration of lift is 33 m/s^2 downward then displacement of the block w.r.t. elevator upto new equilibrium position is 165 cm upward and tension in the cord 1 is 0 N
2. Someone is using a scissors to cut a vertical wire of circular cross section. To reduce required force the wire must be placed close to the hinge; but if it is placed close to the hinge, it slides in the direction away from the hinge until the angle between the scissors blades becomes θ . Find the coefficient of friction between the blades and the wire.



- (A) $\cot(0.5\theta)$ (B) $\tan(0.5\theta)$
- (C) $0.5 \tan \theta$ (D) More information is required to decide.

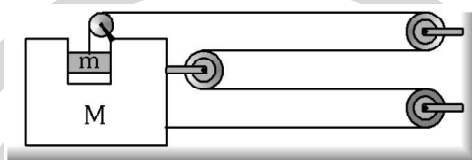
3. A cylinder of mass m rests in a V -shaped block as shown. The maximum acceleration a that the carriage may be given up the incline so that the cylinder does not lose contact at B is (neglect friction) ($g = 10 \text{ m/s}^2$):



- (A) 5.2 m/s^2 (B) 3.66 m/s^2 (C) 2.8 m/s^2 (D) zero

ONE OR MORE THAN ONE OPTION(S) CORRECT: (+4, -1)

4. In the given arrangement string and pulleys are massless and mass of wedge $M = 4m$, then which of the following statements are correct. (Neglect friction everywhere)



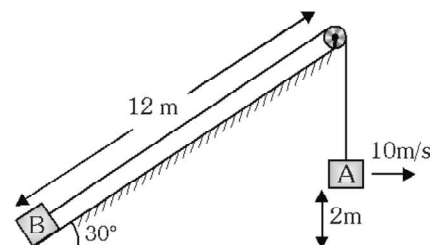
- (A) downward accelerations of m with respect to wedge is $\frac{16g}{21}$
 (B) acceleration of wedge with respect to ground is $\frac{4g}{21}$
 (C) acceleration of mass m with respect to ground $\frac{\sqrt{17}g}{5}$
 (D) acceleration of wedge with respect to ground is $3g/4$
5. In the given arrangement block A is pulled horizontally with constant speed 10 m/s from given position. Block B along smooth inclined plane from given position. (mass of $B = 2 \text{ kg}$)

- (A) Time taken by B to reach at pulleys is $\sqrt{2.4} \text{ sec}$.

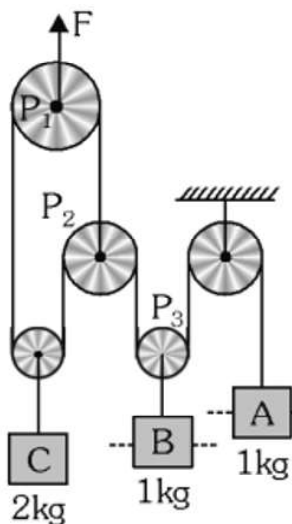
- (B) Acceleration of block B at $t = 1 \text{ sec}$ is $\frac{200}{29\sqrt{29}}$

- (C) Acceleration of block B at $t = 1 \text{ sec}$ is $\frac{100}{29\sqrt{29}}$

- (D) Horizontal component of force of external agent at $t = 1 \text{ sec}$ is $\left(\frac{400}{29\sqrt{29}} + 10 \right) \frac{10}{2\sqrt{29}}$



6. In the given figure all pulleys (other than P_2) and strings are massless. Mass of pulley P_2 is 2 kg . A force $F = 10t$ newton is applied on pulley P_1 vertically up. At $t = 0$ all are at rest [$t =$ time in second] [Assuming parts of string between pulleys or pulley and block are sufficiently large]

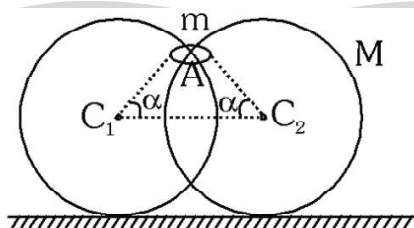


- (A) Acceleration of pulley P_1 at $t = 2$ second is 17.5 m/s^2 upward
 (B) Velocity of block C when acceleration of A is zero is 15 m/s in downward
 (C) If initially block B and C are at same horizontal level, then vertical separation between them when acceleration of block B is zero is $\frac{5}{6}\text{ m}$
 (D) Velocity of block C when acceleration of A is zero is 20 m/s in downward.

PARAGRAPH BASED QUESTION(S) EACH CORRECT: (+3, -1)

Paragraph - 1 for Question Nos. 7 to 9

A small ring of mass m connecting freely two identical thin hoops of mass M each starts sliding down. The hoops move apart over a smooth horizontal surface. The system released from rest when $\angle AC_1C_2 = \alpha$.



7. Initial acceleration of ring is

- (A) $\frac{g}{1 + 2\left(\frac{M}{m}\tan^2 \alpha\right)}$ (B) $\frac{g}{1 + \frac{M}{m}\tan^2 \alpha}$ (C) $\frac{g}{\tan^2 \alpha + \frac{M}{m}}$ (D) None of these

8. Initial acceleration of hoop is

- (A) $\frac{g \tan \alpha}{1 + 2\left(\frac{M}{m} \tan^2 \alpha\right)}$ (B) $\frac{g \tan \alpha}{\left(1 + \frac{M}{m} \tan^2 \alpha\right)}$ (C) $\frac{g \tan \alpha}{\tan^2 \alpha + \frac{M}{m}}$ (D) None of these

9. Normal force between one of the hoop and horizontal surface is.

- (A) equal to $Mg + \frac{mg}{2}$ (B) Greater than $Mg + \frac{mg}{2}$
(C) Less than $Mg + \frac{mg}{2}$ (D) can't predict

NUMERICALS TYPE: (+4, -1)

10. Turns of a uniform spring of relaxed length $\ell = 1.00\text{ m}$ and force constant $k = 500\text{ N/m}$ almost touch each other. A light glue is applied evenly between every adjacent turn. Breaking strength of the glue is $F_b = 100\text{ N}$. The spring is placed on a frictionless horizontal floor and pulled from one of its ends. If the pulling force is gradually increased to a value $F = 200\text{ N}$, how much will the length of the spring become (in meter) ?